Sistemas Distribuídos e Mobile

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Tópicos

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Sistemas Dist

Mãos no código

02

Sistemas Dist

Mãos no código

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Sistemas Dist

Mãos no código

04

Sistemas Dist

Mãos no código

Wow !

Programação deve ser igual banho, diário.

01

Microservice

Sistemas Distribuídos

Criar um novo projeto

Vamos criar um novo projeto utilizando o terminal console! Vamos validar se temos o **SDK** e a **extensão** para desenvolver:

PowerShell

```
C:\Users\paiva

> dotnet --version
10.0.102

C:\Users\paiva

>
```

Extension: C# Dev Kit

C# Dev Kit

Microsoft microsoft.com | 14,217,795 | (194)

Official C# extension from Microsoft

Switch to Pre-Release Version | Disable | Uninstall | Auto Update

DETAILS FEATURES CHANGELOG DEPENDENCIES EXTENSION PACK

C# Dev Kit for Visual Studio Code

C# Dev Kit helps you manage your code with a solution explorer and test your code with integrated unit test discovery and execution, elevating your C# development experience wherever you like to develop (Windows, macOS, Linux, and even in a Codespace).

This extension builds on top of the great C# language capabilities provided by the C# extension and enhances your C# environment by adding a set of powerful tools and utilities that integrate natively with VS Code to help C# developers write, debug, and maintain their code faster and with fewer errors. Some of this new tooling includes but is not limited to:

- C# project and solution management via an integrated solution explorer
- Native testing environment to run and debug tests using the Test Explorer
- Roslyn-powered language service for best in-class C# language features such as code navigation, refactoring, semantic awareness, and more

Quick Start

Installation

Identifier	ms-dotnettools.csdevkit
Version	2.10.3
Last Updated	1 month ago
Size	167.47MB

Marketplace

Published	3 years ago
Last Released	3 weeks ago

Categories

Programming Languages | Debuggers | Testing | Linters

Extension: C# Extensions

C# Extensions

JosKreativ | 1,419,126 | (15)

C# IDE Extensions for VSCode

Disable | Uninstall | Auto Update

DETAILS FEATURES CHANGELOG

Customized version of https://github.com/jchannon/csharpextensions, -> https://github.com/KreativJos/csharpextensions

C# Extensions

Welcome to C# Extensions. This VSCode extension provides extensions to the IDE that will hopefully speed up your development workflow. It can currently be found at:

Installation

Identifier	kreativ-software.csharpextensions
Version	1.7.3
Last Updated	5 months ago
Size	239.16KB

Marketplace

Published	7 years ago

Criar um novo projeto

Vamos criar um novo projeto utilizando o terminal console!
Para isso, abra o Visual Studio Code já na pasta que você criará o projeto OlaMundoApi.

```
dotnet new webapi -n NomeDaSuaApi --use-controllers
```

PowerShell

```
C:\Users\paiva

> dotnet new webapi -n OlaMundoApi --use-controllers
```

PowerShell

```
> dotnet new webapi -n OlaMundoApi --use-controllers
O modelo "API Web do ASP.NET Core" foi criado com êxito.

Processando ações pós-criação...
Restaurando C:\Users\paiva\OlaMundoApi\OlaMundoApi.csproj:
A restauração foi bem-sucedida.



C:\Users\paiva

> cd .\OlaMundoApi\

C:\Users\paiva\OlaMundoApi

>
```

PowerShell

```
Processando ações pós-criação...
Restaurando C:\Users\paiva\OlaMundoApi\OlaMundoApi.csproj:
A restauração foi bem-sucedida.



C:\Users\paiva

> cd .\OlaMundoApi\

C:\Users\paiva\OlaMundoApi

> code .

C:\Users\paiva\OlaMundoApi

>
```

EXPLORER

OLAMUNDOAPI

- Controllers
- obj
- Properties
- appsettings.Development.json
- appsettings.json
- OlaMundoApi.csproj
- OlaMundoApi.http
- Program.cs
- WeatherForecast.cs

OlaMundoApi - WeatherForecast.cs

```
1   namespace OlaMundoApi;
2
3   public class WeatherForecast
4   {
5       public DateOnly Date { get; set; }
6
7       public int TemperatureC { get; set; }
8
9       public int TemperatureF => 32 + (int)(TemperatureC / 0.5556);
10
11      public string? Summary { get; set; }
12  }
13
```

OlaMundoApi - WeatherForecastController.cs

```
using Microsoft.AspNetCore.Mvc;

namespace OlaMundoApi.Controllers;

[ApiController]
[Route("[controller]")]
public class WeatherForecastController : ControllerBase
{
    private static readonly string[] Summaries =
    [
        "Freezing", "Bracing", "Chilly", "Cool", "Mild", "Warm", "Balmy", "Hot", "Sweltering", "Scorching"
    ];

    [HttpGet(Name = "GetWeatherForecast")]
    public IEnumerable<WeatherForecast> Get()
    {
        return Enumerable.Range(1, 5).Select(index => new WeatherForecast
        {
            Date = DateOnly.FromDateTime(DateTime.Now.AddDays(index)),
            TemperatureC = Random.Shared.Next(-20, 55),
            Summary = Summaries[Random.Shared.Next(Summaries.Length)]
        })
        .ToArray();
    }
}
```

OlaMundoApi - launchSettings.json

```json
{
  "$schema": "https://json.schemastore.org/launchsettings.json",
  "profiles": {
    "http": {
      "commandName": "Project",
      "dotnetRunMessages": true,
      "launchBrowser": false,
      "applicationUrl": "http://localhost:5258",
      "environmentVariables": {
        "ASPNETCORE_ENVIRONMENT": "Development"
      }
    },
    "https": {
      "commandName": "Project",
      "dotnetRunMessages": true,
      "launchBrowser": false,
      "applicationUrl": "https://localhost:7041;http://localhost:5258",
      "environmentVariables": {
        "ASPNETCORE_ENVIRONMENT": "Development"
      }
    }
  }
}
```

OlaMundoApi - Program.cs

```
var builder = WebApplication.CreateBuilder(args);

// Add services to the container.

builder.Services.AddControllers();
// Learn more about configuring OpenAPI at https://aka.ms/aspnet/openapi
builder.Services.AddOpenApi();

var app = builder.Build();

// Configure the HTTP request pipeline.
if (app.Environment.IsDevelopment())
{
    app.MapOpenApi();
}

app.UseHttpsRedirection();

app.UseAuthorization();

app.MapControllers();

app.Run();

```

Windows PowerShell

```
PS C:\Users\paiva\Documents\OlaMundoApi> dotnet build
Restauração concluída (0,4s)
  OlaMundoApi net10.0 êxito (7,4s) → bin\Debug\net10.0\OlaMundoApi.dll

Construir êxito em 8,2s
PS C:\Users\paiva\Documents\OlaMundoApi>
```

Windows PowerShell

```
Construir êxito em 8,2s
PS C:\Users\paiva\Documents\OlaMundoApi> dotnet run
Usando as configurações de inicialização de C:\Users\paiva\Documents\OlaMundoApi\Pr
operties\launchSettings.json...
Compilando...
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://localhost:5258
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: C:\Users\paiva\Documents\OlaMundoApi
```

Windows PowerShell

```
Construir êxito em 8,2s
PS C:\Users\paiva\Documents\OlaMundoApi> dotnet run
Usando as configurações de inicialização de C:\Users\paiva\Documents\OlaMundoApi\Pr
operties\launchSettings.json...
Compilando...
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://localhost:5258
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: C:\Users\paiva\Documents\OlaMundoApi
```

localhost

localhost:5258

Não foi possível encontrar a página deste localhost

Nenhuma página da web foi encontrada para o endereço da Web:**http://localhost:5258/**

HTTP ERROR 404

Recarregar

WeatherForecastController.cs

```
using Microsoft.AspNetCore.Mvc;

namespace OlaMundoApi.Controllers;

[ApiController]
[Route("[controller]")]
public class WeatherForecastController : ControllerBase
{
    private static readonly string[] Summaries =
    [
        "Freezing", "Bracing", "Chilly", "Cool", "Mild", "Warm", "Balmy", "Hot", "Sweltering", "Scorching"
    ];

    [HttpGet(Name = "GetWeatherForecast")]
    public IEnumerable<WeatherForecast> Get()
    {
        return Enumerable.Range(1, 5).Select(index => new WeatherForecast
        {
            Date = DateOnly.FromDateTime(DateTime.Now.AddDays(index)),
            TemperatureC = Random.Shared.Next(-20, 55),
            Summary = Summaries[Random.Shared.Next(Summaries.Length)]
        })
        .ToArray();
    }
}
```

ENDPOINT

http://localhost:5258/weatherforecast

WeatherForecastController.cs

```
1  using Microsoft.AspNetCore.Mvc;
2
3  namespace OlaMundoApi.Controllers;
4
5  [ApiController]
6  [Route("[controller]")]
7  public class WeatherForecastController :
8  {
9      private static readonly string[] Summ
```

localhost:5258/weatherforecast

localhost:5258/weatherforecast

Estilos de formatação

[{"date":"2026-02-17","temperatureC":54,"temperatureF":129,"summary":"Sweltering"},{"date":"2026-02-18","temperatureC":-16,"temperatureF":4,"summary":"Freezing"},{"date":"2026-02-19","temperatureC":4,"temperatureF":39,"summary":"Hot"},{"date":"2026-02-20","temperatureC":16,"temperatureF":60,"summary":"Mild"},{"date":"2026-02-21","temperatureC":45,"temperatureF":112,"summary":"Hot"}]

Vamos criar uma nova controller

Vamos criar uma nova controller chamada OlaMundoController.

EXPLORER

OLAMUNDOAPI

- bin
- Controllers
  - OlaMundoController.cs
  - WeatherForecastController.cs
- obj
- Properties
  - launchSettings.json
- appsettings.Development.json
- appsettings.json
- OlaMundoApi.csproj
- OlaMundoApi.http
- Program.cs
- WeatherForecast.cs

OlaMundoApi - OlaMundoController.cs

```
1   using System;
2   using System.Collections.Generic;
3   using System.Linq;
4   using System.Threading.Tasks;
5
6   namespace OlaMundoApi.Controllers
7   {
8       public class OlaMundoController
9       {
10
11      }
12  }
```

Vamos criar uma nova controller

Vamos decorar a classe...

OlaMundoApi - OlaMundoController.cs

```
1   using System;
2   using System.Collections.Generic;
3   using System.Linq;
4   using System.Threading.Tasks;
5
6   namespace OlaMundoApi.Controllers
7   {
8       [ApiController]
9       [Route("[controller]")]
10      public class OlaMundoController : ControllerBase
11      {
12
13      }
14  }
```

Vamos criar uma nova controller

Vamos criar o método que retorna uma string “Ola, Mundo!”...

OlaMundoApi - OlaMundoController.cs

```
1  [ApiController]
2  [Route("[controller]")]
3  public class OlaMundoController : ControllerBase
4  {
5      public string Get()
6      {
7          return "Olá, Mundo!";
8      }
9  }
```

OlaMundoApi - OlaMundoController.cs

```
1  using System;
2  using System.Collections.Generic;
3  using System.Linq;
4  using System.Threading.Tasks;
5  using Microsoft.AspNetCore.Mvc;
6
7  namespace OlaMundoApi.Controllers
8  {
9      [ApiController]
10     [Route("[controller]")]
11     public class OlaMundoController : ControllerBase
12     {
13         public string Get()
14         {
15             return "Olá, Mundo!";
16         }
17     }
18 }
```

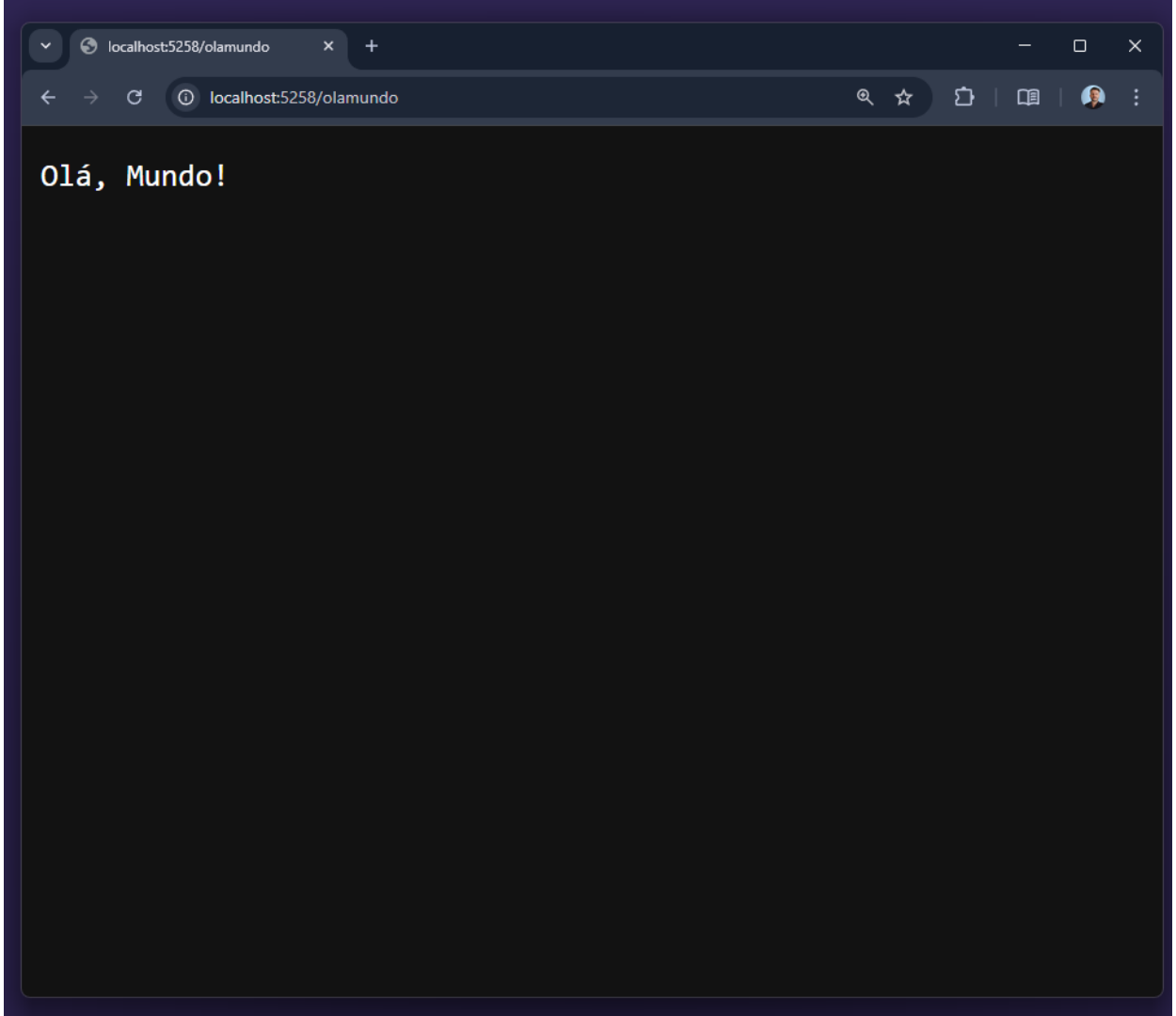
Vamos criar uma nova controller

Vamos testar a nossa nova controller...

Windows PowerShell

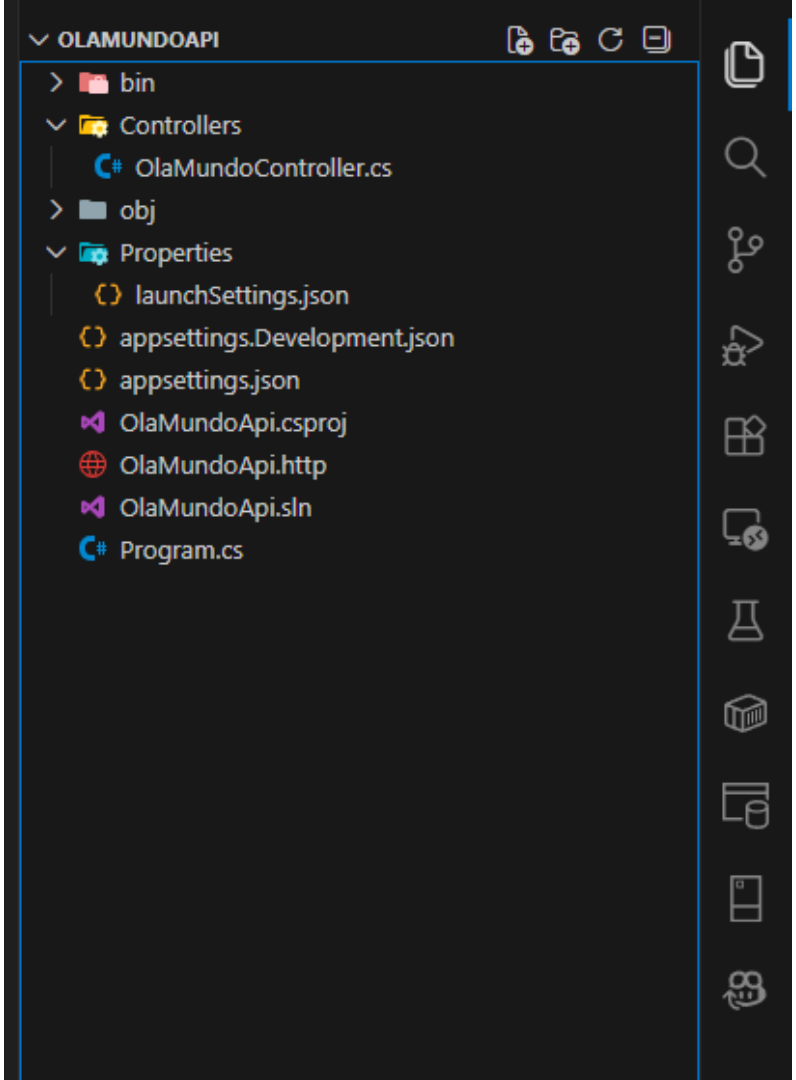
```
PS C:\Users\paiva\Documents\OlaMundoApi> dotnet build
Restauração concluída (0,2s)
  OlaMundoApi net10.0 êxito (0,3s) → bin\Debug\net10.0\OlaMundoApi.dll

Construir êxito em 0,9s
PS C:\Users\paiva\Documents\OlaMundoApi> dotnet run
Usando as configurações de inicialização de C:\Users\paiva\Documents\OlaM
undoApi\Properties\launchSettings.json...
Compilando...
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://localhost:5258
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: C:\Users\paiva\Documents\OlaMundoApi
```


localhost:5258/olamundo
localhost:5258/olamundo
Olá, Mundo!

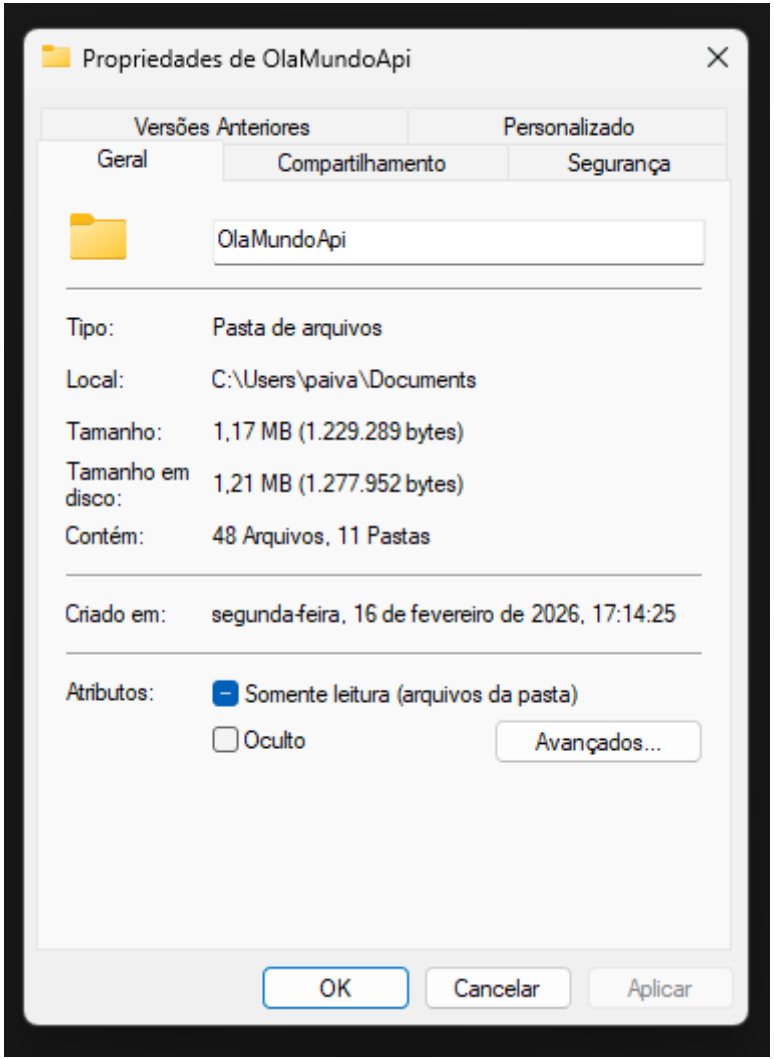
Limpando o projeto

Podemos excluir a classe **WeatherForecast** e **WeatherForecastController** que não pertence ao contexto da nossa API.


OLAMUNDOAPI
bin
Controllers
OlaMundoController.cs
obj
Properties
launchSettings.json
appsettings.Development.json
appsettings.json
OlaMundoApi.csproj
OlaMundoApi.http
OlaMundoApi.sln
Program.cs

Limpando o projeto

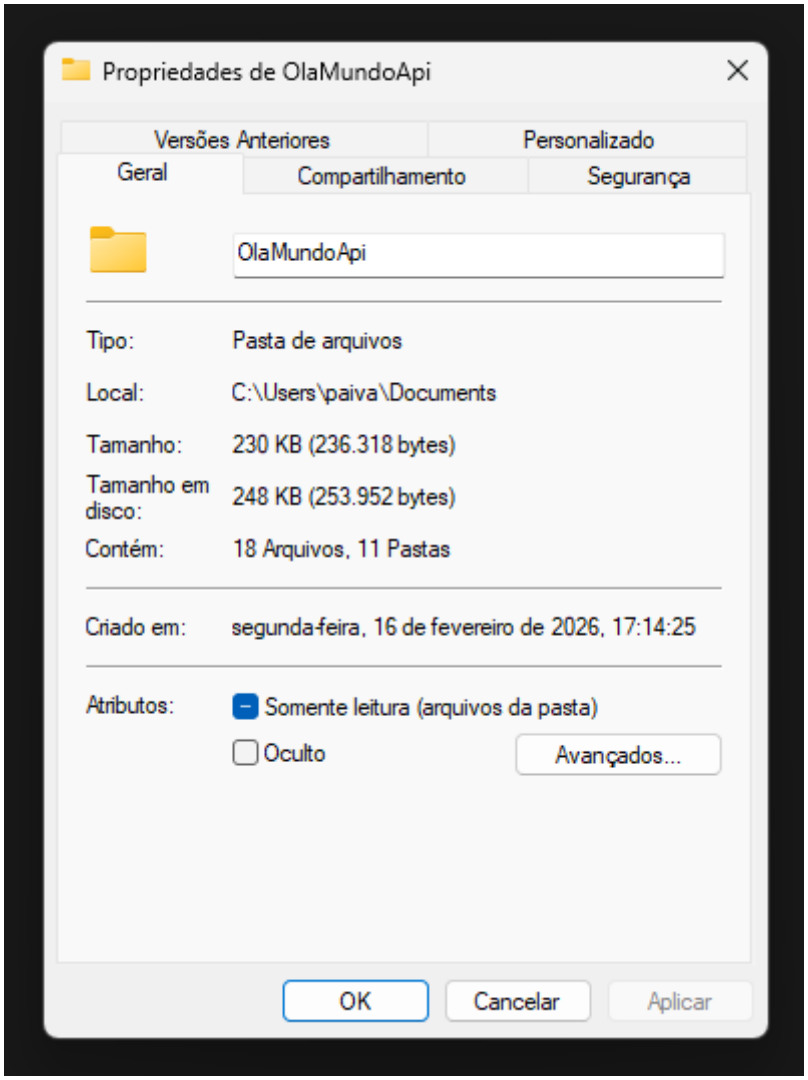
Vamos limpar o nosso projeto com o comando:
dotnet clean


Propriedades de OlaMundoApi
Versões Anteriores
Personalizado
Geral
Compartilhamento
Segurança
OlaMundoApi
Tipo:
Pasta de arquivos
Local:
C:\Users\paiva\Documents
Tamanho:
1,17 MB (1.229.289 bytes)
Tamanho em disco:
1,21 MB (1.277.952 bytes)
Contém:
48 Arquivos, 11 Pastas
Criado em:
segunda-feira, 16 de fevereiro de 2026, 17:14:25
Atributos:
Somente leitura (arquivos da pasta)
Oculto
Avançados...
OK
Cancelar
Aplicar

Windows PowerShell

```
PS C:\Users\paiva\Documents\OlaMundoApi> dotnet clean

Construir êxito em 0,4s
PS C:\Users\paiva\Documents\OlaMundoApi>
```


Propriedades de OlaMundoApi
Versões Anteriores
Personalizado
Geral
Compartilhamento
Segurança
OlaMundoApi
Tipo:
Pasta de arquivos
Local:
C:\Users\paiva\Documents
Tamanho:
230 KB (236.318 bytes)
Tamanho em disco:
248 KB (253.952 bytes)
Contém:
18 Arquivos, 11 Pastas
Criado em:
segunda-feira, 16 de fevereiro de 2026, 17:14:25
Atributos:
Somente leitura (arquivos da pasta)
Oculto
Avançados...
OK
Cancelar
Aplicar

Vamos criar uma nova controller

Agora suba sua nova api para seu GitHub pessoal e envie o link do repositório.
Nome: **una-olamundoapi-net**

Até breve!

Dúvidas???

profdanhpaiva@gmail.com
https://www.linkedin.com/in/danhpaiva/